

# Evaluation Metrics Report

Data Vortex 2026 · Round 2 · Rebuilding the Semantic Layer

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Dataset 2: 7,900 modeled rows (train 6,320 / test 1,580, seed 42)

Sentiment: logreg\_union -- test macro-F1 0.6134 · Topic: svc\_union -- test macro-F1  
0.8051

Every number in this report is computed by the Round 2 pipeline from Dataset 2 and the trained models -- nothing is typed by hand. Re-run ./run\_round2.sh to reproduce this PDF byte-for-byte (modulo timestamps).

## Dataset & split

Row reconciliation: 9000 raw - 0 empty - 1100 exact-duplicate = 7900 modeled. Empty-text rows (0) and exact (text, labels) duplicates (1,100) are held out in named files under data/round2/clean/ -- never silently dropped, never imputed. 0 distinct texts carry conflicting labels across rows and are kept as-is (reported label noise).

| sentiment | rows  | share |
|-----------|-------|-------|
| Negative  | 2,430 | 30.8% |
| Neutral   | 2,748 | 34.8% |
| Positive  | 2,722 | 34.5% |

| topic                | rows  | share |
|----------------------|-------|-------|
| Community_Discussion | 6,803 | 86.1% |
| Technical_Issues     | 718   | 9.1%  |
| Feature_Feedback     | 250   | 3.2%  |
| Account_Security     | 129   | 1.6%  |

Split: 6,320 train / 1,580 test (stratified: sentiment\_x\_topic, seed 42, id overlap 0). Text length: median 112 chars, mean 106.5, max 158.

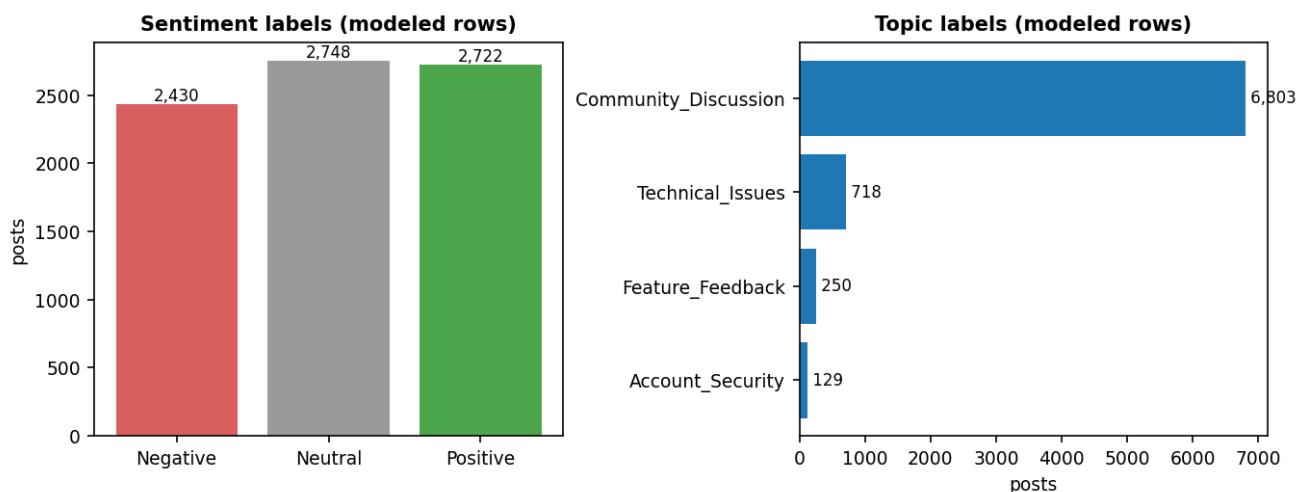


Figure: label distributions over modeled rows.

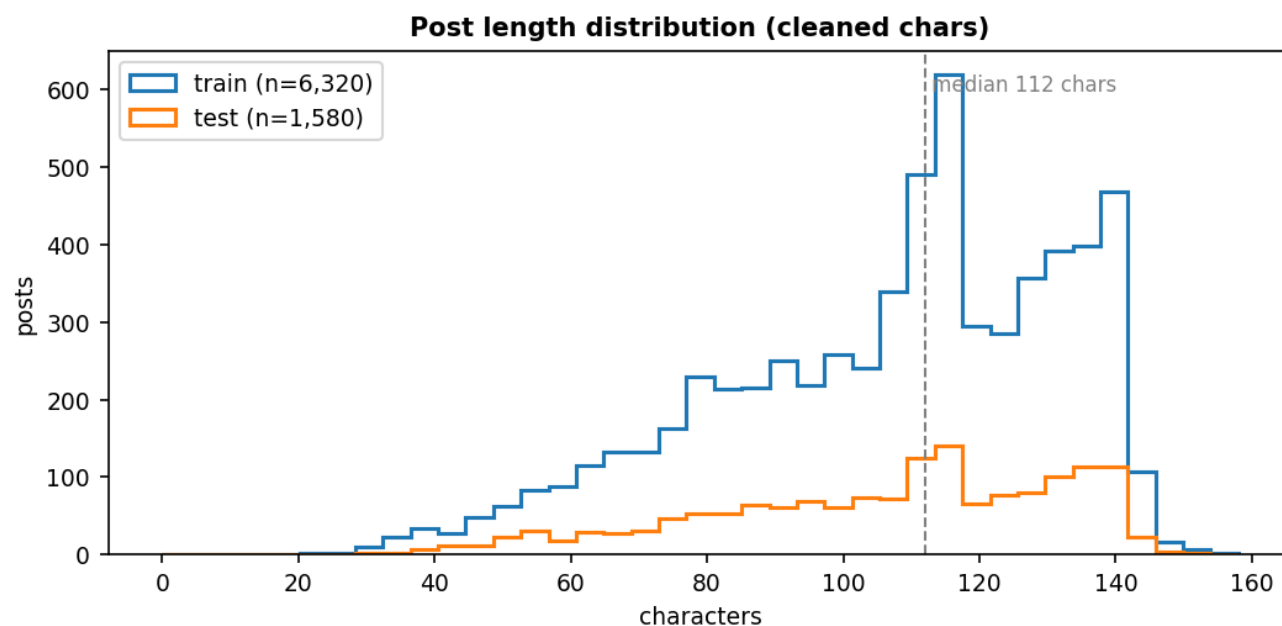


Figure: post-length distributions -- train vs test overlap shows the split did not segregate short/long posts.

## Task: sentiment classification

### Model selection (CV on train only)

5-fold stratified cross-validation, scoring macro-F1. Winner: logreg\_union (CV  $0.6110 \pm 0.0148$ ), margin +0.0081 over runner-up svc\_union. Winner ships uncalibrated (native probabilities).

| model          | CV macro-F1 (mean $\pm$ sd) | best params                   | fit (s) |
|----------------|-----------------------------|-------------------------------|---------|
| logreg_union ★ | $0.6110 \pm 0.0148$         | C=0.5, class_weight=balanced  | 59.1    |
| svc_union      | $0.6028 \pm 0.0091$         | C=0.25, class_weight=None     | 37.3    |
| logreg_word    | $0.5915 \pm 0.0156$         | C=1.0, class_weight=balanced  | 13.2    |
| svc_word       | $0.5893 \pm 0.0136$         | C=0.25, class_weight=balanced | 9.4     |
| nb_word        | $0.5709 \pm 0.0120$         | alpha=0.5                     | 3.7     |
| dummy          | $0.3240 \pm 0.0067$         | —                             | 1.2     |

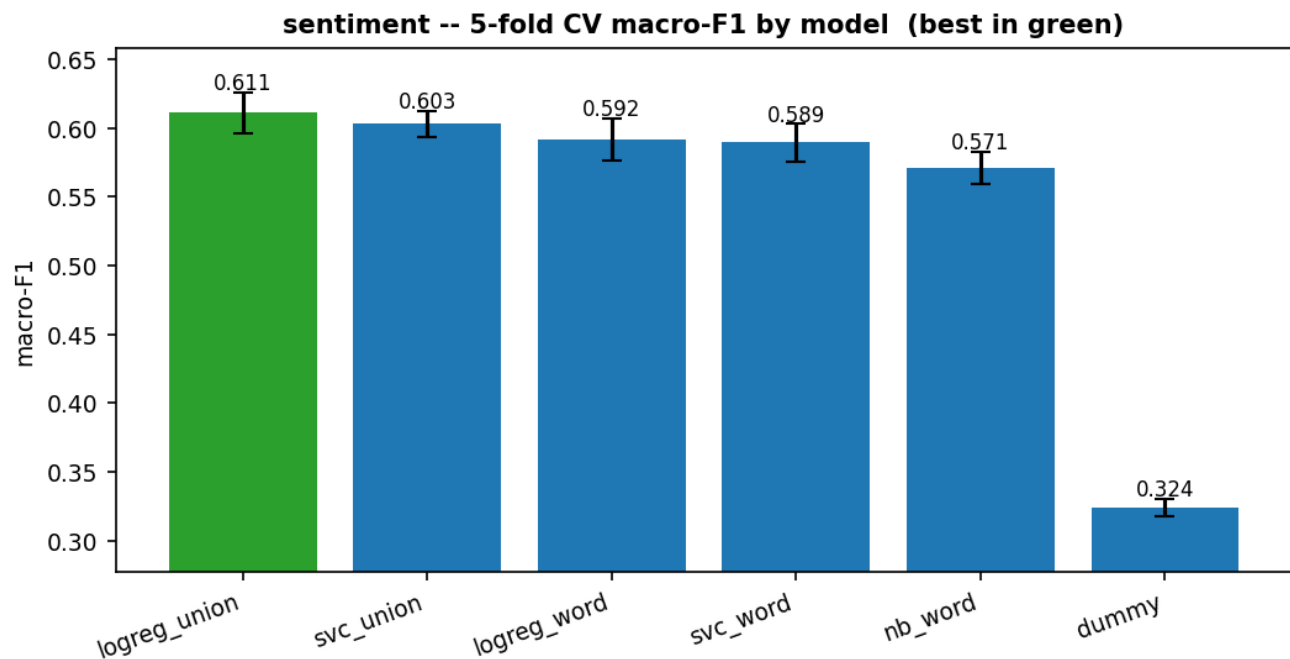


Figure: sentiment -- mean CV macro-F1 per model (error bar =  $\pm 1$  sd across folds; winner in green).

Head-to-head on the test set (logreg\_union (shipped) vs svc\_union (runner-up)): b=68 rows where shipped wins, c=51 where runner-up wins, McNemar exact  $p=0.1421$  -- not significant at 5%: the lead is within chance.

### Held-out test metrics

| metric             | value  |
|--------------------|--------|
| accuracy           | 0.6133 |
| macro-F1 (primary) | 0.6134 |
| weighted-F1        | 0.6117 |
| micro-F1           | 0.6133 |
| MCC                | 0.4201 |
| ROC-AUC (OvR)      | 0.7916 |
| log-loss           | 0.8621 |
| Brier (OvR mean)   | 0.1703 |
| test rows          | 1,580  |

| class    | precision | recall | F1     | support |
|----------|-----------|--------|--------|---------|
| Negative | 0.6337    | 0.6728 | 0.6527 | 486     |
| Neutral  | 0.5655    | 0.5191 | 0.5413 | 549     |

| class    | precision | recall | F1     | support |
|----------|-----------|--------|--------|---------|
| Positive | 0.6375    | 0.6550 | 0.6462 | 545     |

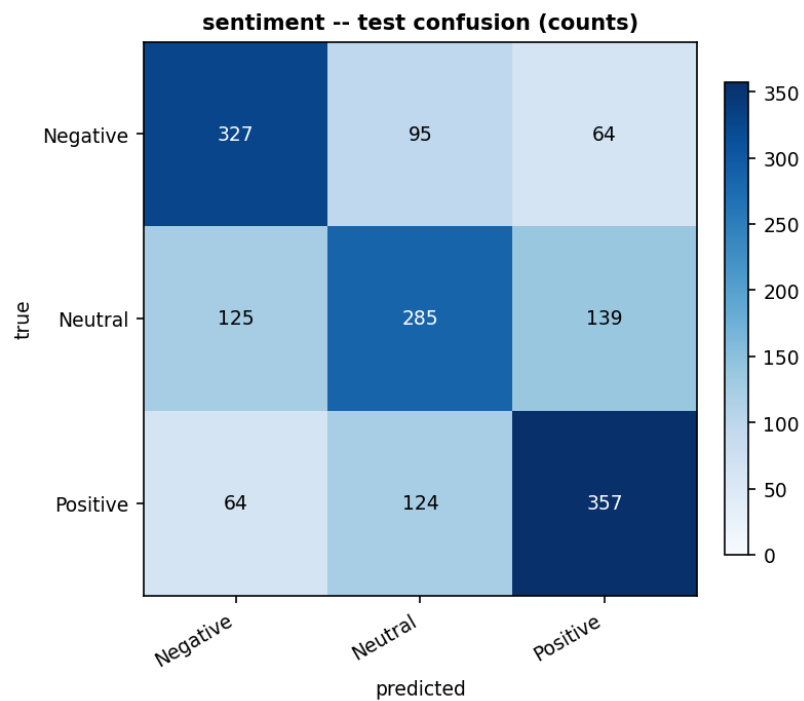


Figure: sentiment -- test confusion matrix (counts).

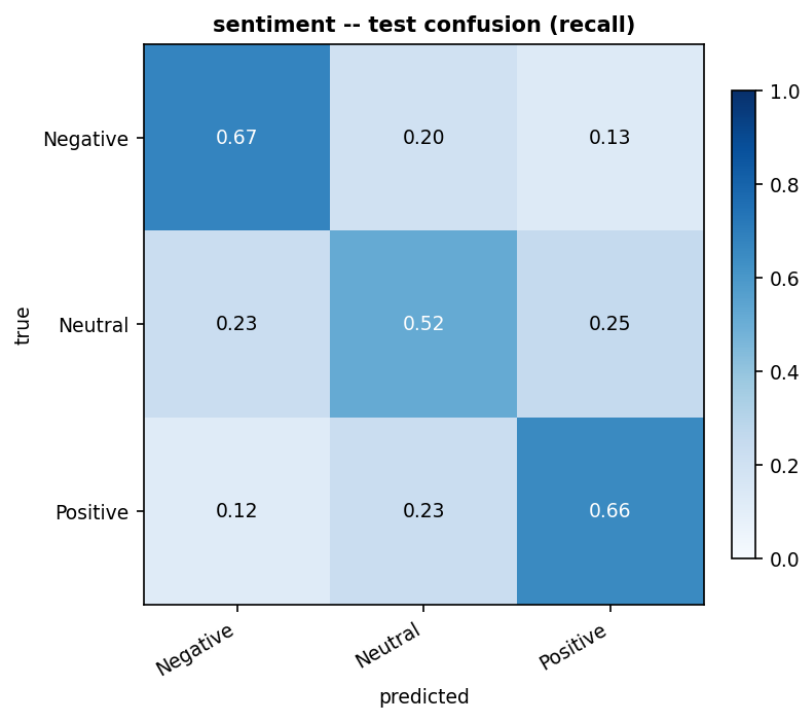


Figure: sentiment -- test confusion matrix (row-normalised = recall).

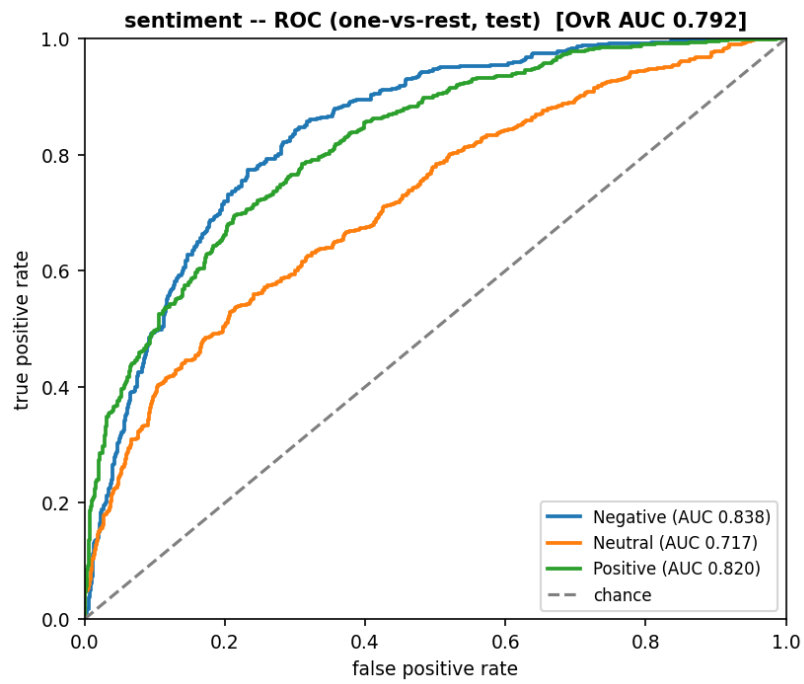


Figure: sentiment -- ROC curves, one-vs-rest (test).

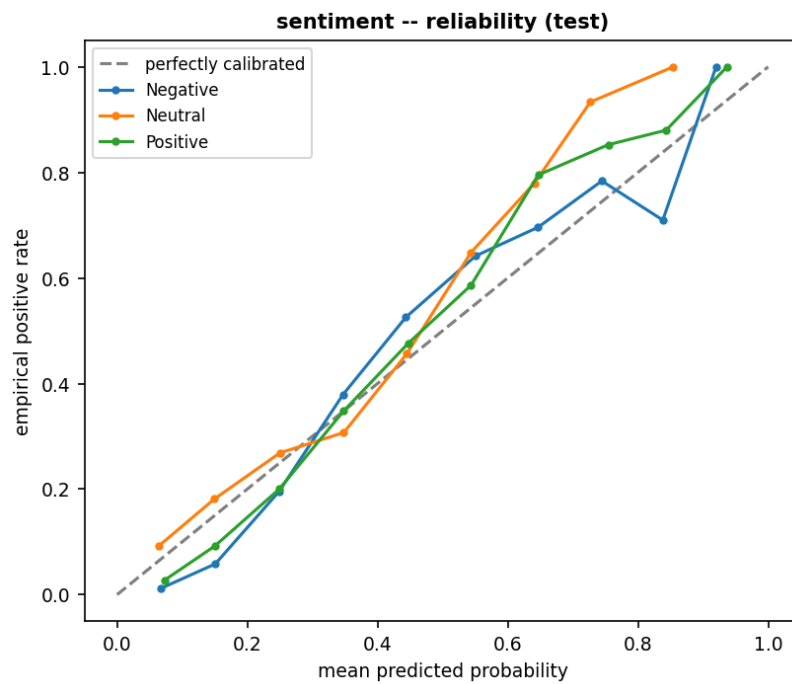


Figure: sentiment -- reliability diagram (test). Points on the diagonal = predicted probabilities match empirical rates.

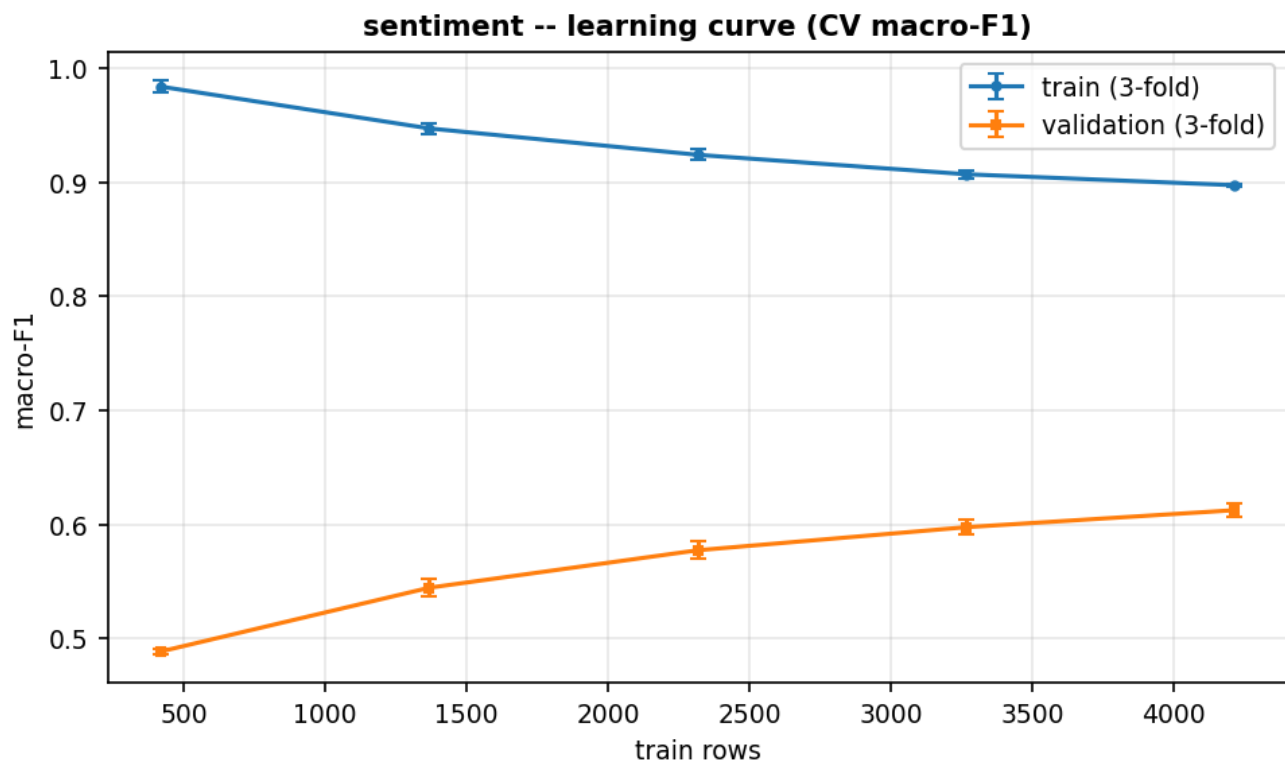


Figure: sentiment -- learning curve (3-fold CV macro-F1 vs train size). A plateauing validation curve means more labels would barely help.

### Slice analysis (where it fails)

Accuracy and macro-F1 on interpretable slices of the test set. Slices are computed from the raw text only (length quartile, negation / emoji / URL presence) -- no label leakage.

| slice         | n     | accuracy | macro-F1 |
|---------------|-------|----------|----------|
| overall       | 1,580 | 0.6133   | 0.6134   |
| len_short     | 494   | 0.5709   | 0.5697   |
| len_medium    | 341   | 0.5836   | 0.5852   |
| len_long      | 357   | 0.6891   | 0.6791   |
| len_very_long | 388   | 0.6237   | 0.6233   |
| negation_yes  | 245   | 0.6082   | 0.5325   |
| negation_no   | 1,335 | 0.6142   | 0.6152   |
| emoji_yes     | 61    | 0.7541   | 0.5930   |
| emoji_no      | 1,519 | 0.6076   | 0.6086   |
| url_yes       | 2     | 0.5000   | 0.2222   |
| url_no        | 1,578 | 0.6134   | 0.6135   |

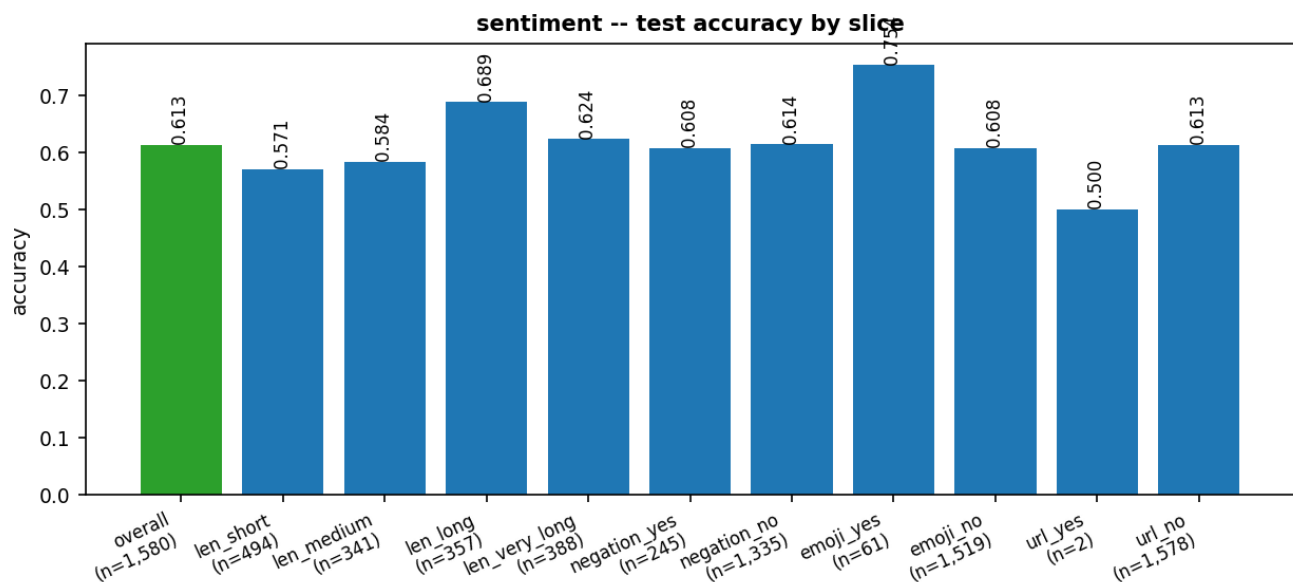


Figure: sentiment -- test accuracy by slice.

## Most confused pairs

| true → predicted    | rows |
|---------------------|------|
| Neutral → Positive  | 139  |
| Neutral → Negative  | 125  |
| Positive → Neutral  | 124  |
| Negative → Neutral  | 95   |
| Positive → Negative | 64   |

## Task: topic classification

### Model selection (CV on train only)

5-fold stratified cross-validation, scoring macro-F1. Winner: svc\_union (CV  $0.7800 \pm 0.0374$ ), margin +0.0134 over runner-up logreg\_union. Winner ships sigmoid-calibrated (LinearSVC has no native probabilities).

| model        | CV macro-F1 (mean $\pm$ sd) | best params                  | fit (s) |
|--------------|-----------------------------|------------------------------|---------|
| svc_union ★  | $0.7800 \pm 0.0374$         | C=2.0, class_weight=balanced | 39.0    |
| logreg_union | $0.7666 \pm 0.0339$         | C=1.0, class_weight=balanced | 58.0    |
| svc_word     | $0.6042 \pm 0.0361$         | C=2.0, class_weight=balanced | 9.6     |
| logreg_word  | $0.5791 \pm 0.0278$         | C=4.0, class_weight=balanced | 14.4    |
| nb_word      | $0.3768 \pm 0.0452$         | alpha=0.1                    | 3.9     |
| dummy        | $0.2397 \pm 0.0112$         | —                            | 0.1     |

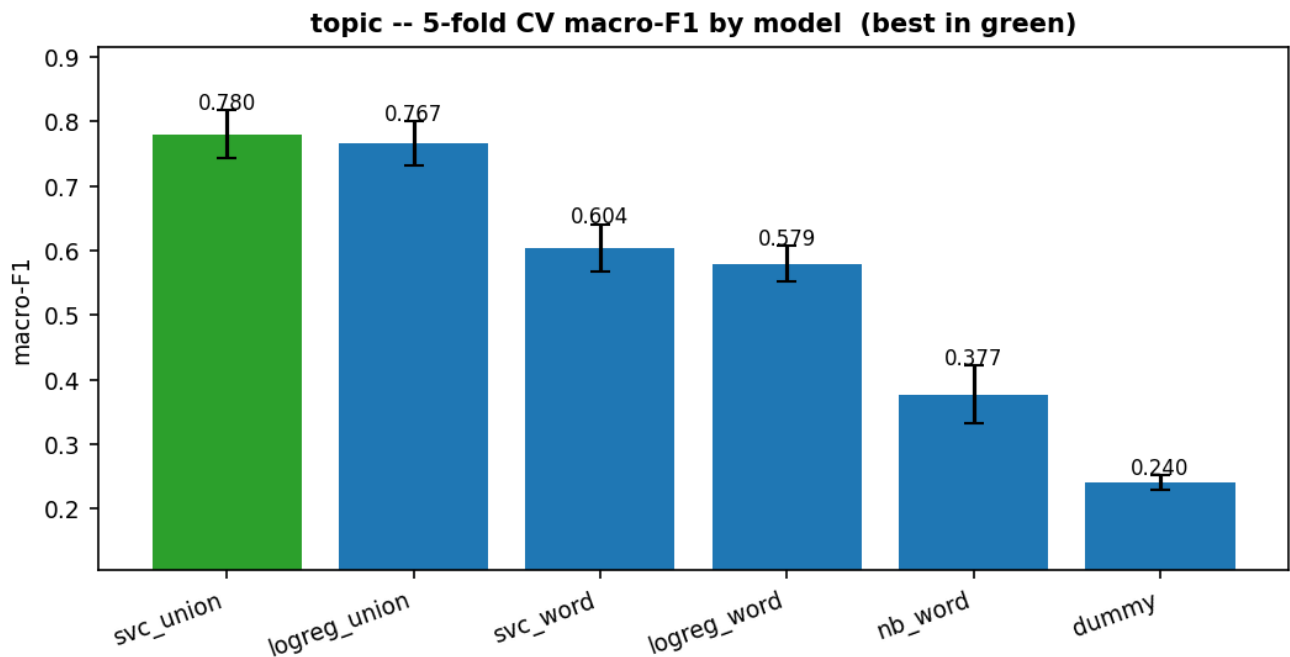


Figure: topic -- mean CV macro-F1 per model (error bar =  $\pm 1$  sd across folds; winner in green).

Head-to-head on the test set (svc\_union\* (shipped) vs logreg\_union (runner-up)): b=23 rows where shipped wins, c=1 where runner-up wins, McNemar exact  $p=0.0000$  -- significant at 5%: the gap is real.

## Held-out test metrics

| metric             | value  |
|--------------------|--------|
| accuracy           | 0.9677 |
| macro-F1 (primary) | 0.8051 |
| weighted-F1        | 0.9631 |
| micro-F1           | 0.9677 |
| MCC                | 0.8645 |
| ROC-AUC (OvR)      | 0.9789 |
| log-loss           | 0.1163 |
| Brier (OvR mean)   | 0.0140 |
| test rows          | 1,580  |

| class                | precision | recall | F1     | support |
|----------------------|-----------|--------|--------|---------|
| Account_Security     | 1.0000    | 0.5000 | 0.6667 | 26      |
| Community_Discussion | 0.9652    | 0.9993 | 0.9819 | 1,360   |
| Feature_Feedback     | 1.0000    | 0.4400 | 0.6111 | 50      |
| Technical_Issues     | 0.9854    | 0.9375 | 0.9609 | 144     |

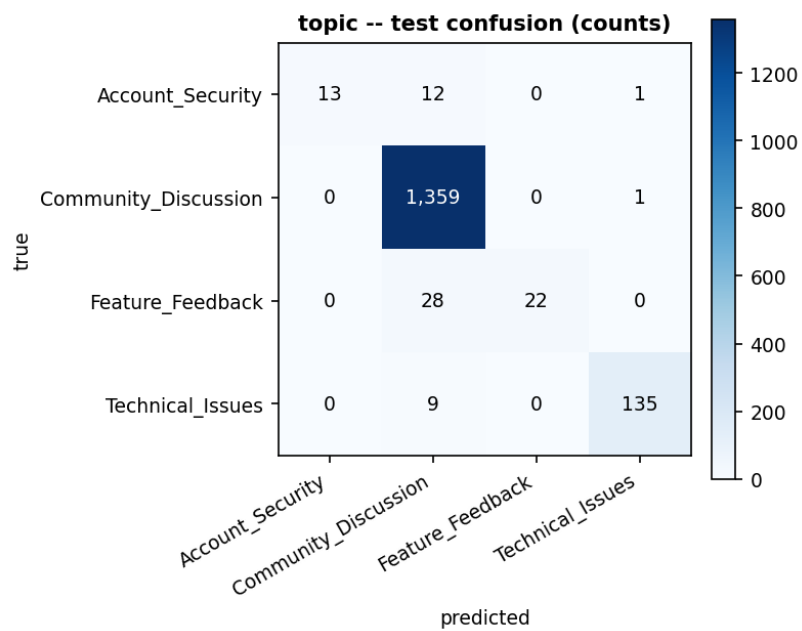


Figure: topic -- test confusion matrix (counts).

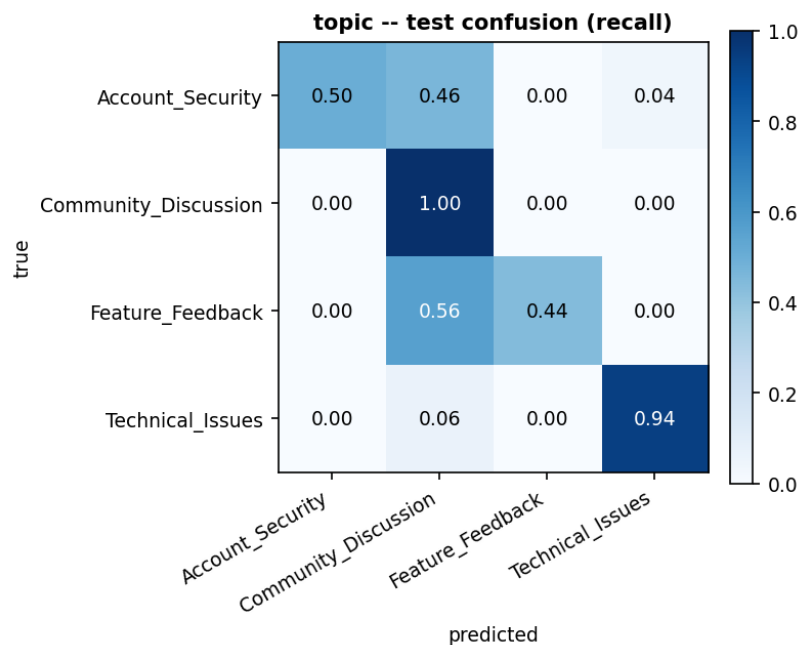


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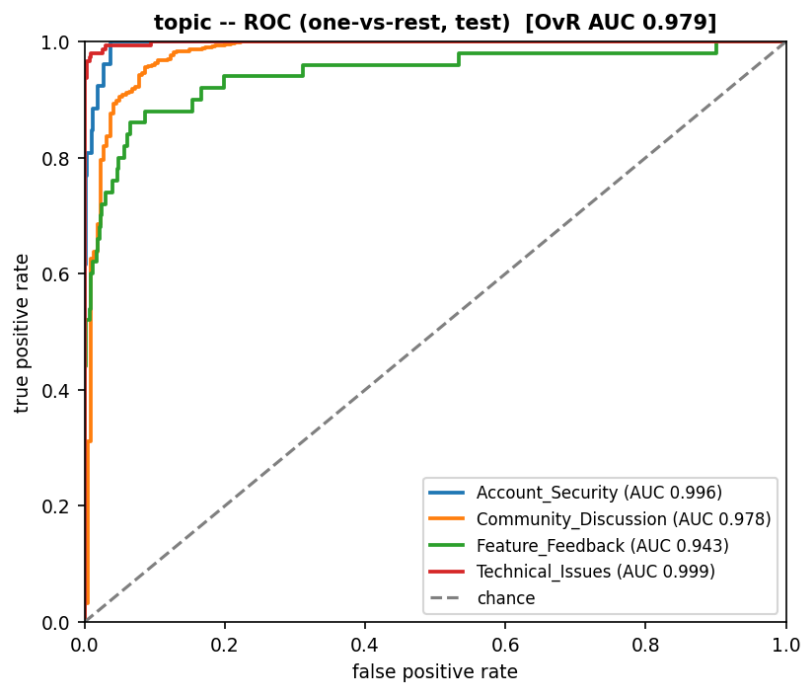


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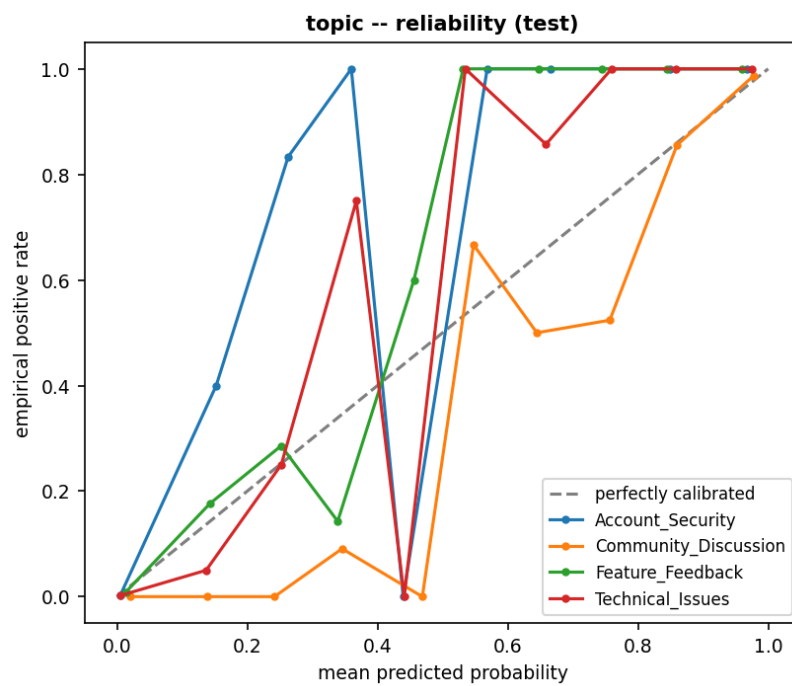


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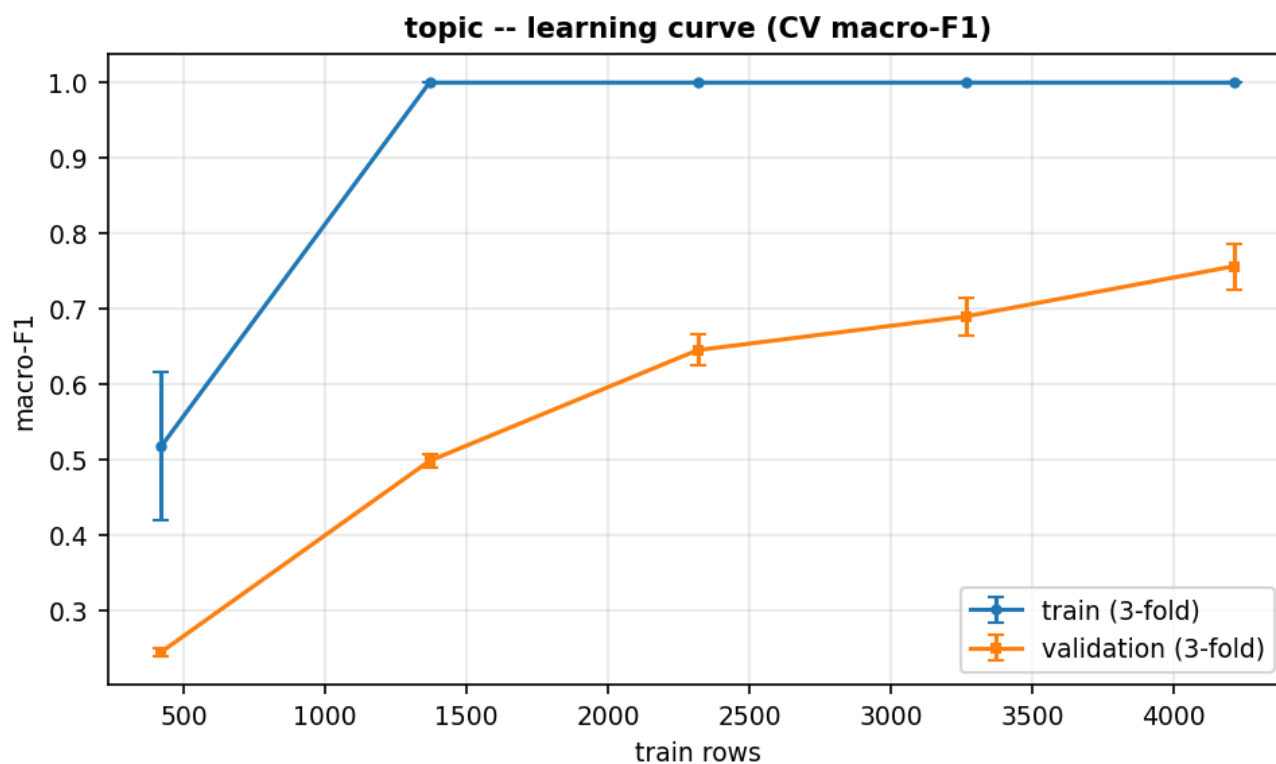


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| negation_yes  | 245   | 0.9755   | 0.8894   |
| negation_no   | 1,335 | 0.9663   | 0.8011   |
| emoji_yes     | 61    | 0.9672   | 0.6620   |
| emoji_no      | 1,519 | 0.9677   | 0.8086   |
| url_yes       | 2     | 1.0000   | 0.5000   |
| url_no        | 1,578 | 0.9677   | 0.8024   |

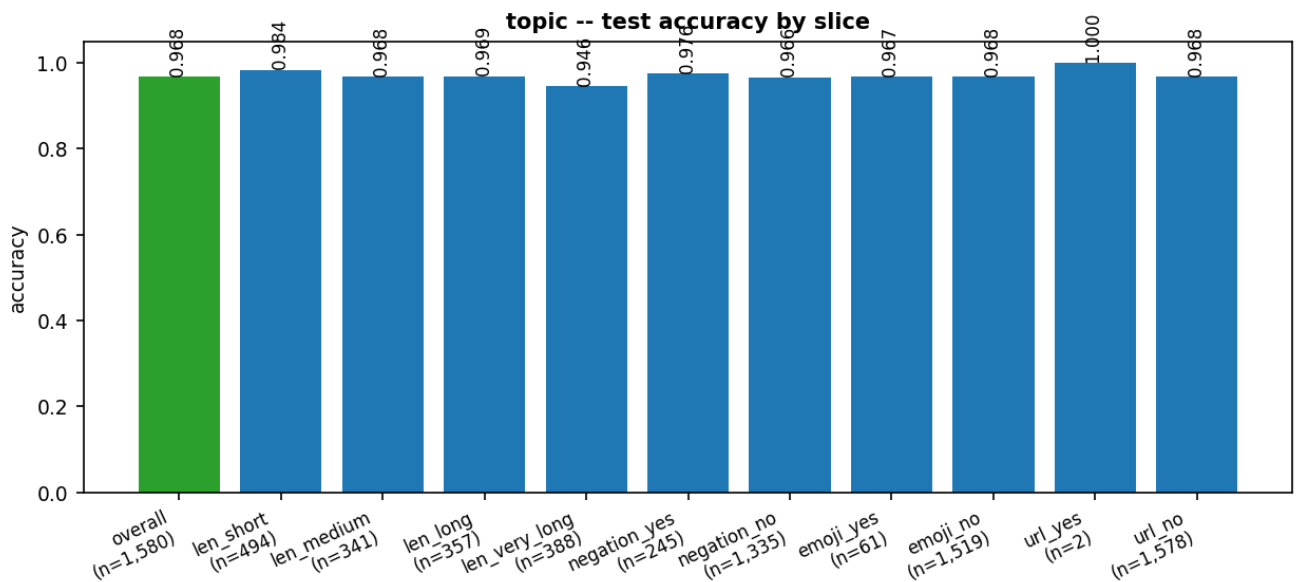


Figure: topic -- test accuracy by slice.

## Most confused pairs

| true → predicted                        | rows |
|-----------------------------------------|------|
| Feature_Feedback → Community_Discussion | 28   |
| Account_Security → Community_Discussion | 12   |
| Technical_Issues → Community_Discussion | 9    |
| Community_Discussion → Technical_Issues | 1    |
| Account_Security → Technical_Issues     | 1    |

## Unsupervised topic-model agreement (bonus)

LDA and NMF, K=4 (= number of human topic labels), fit on train texts without labels; agreement scored on TEST texts by comparing each post's argmax topic against its human topic label.

| model | NMI    | ARI     | homogeneity | completeness | fit quality       |
|-------|--------|---------|-------------|--------------|-------------------|
| LDA   | 0.0020 | -0.0030 | 0.0035      | 0.0014       | perplexity 2609.0 |
| NMF   | 0.0027 | 0.0068  | 0.0046      | 0.0019       | recon err 231.2   |

Unsupervised-vs-human agreement (row-normalised, counts in brackets)

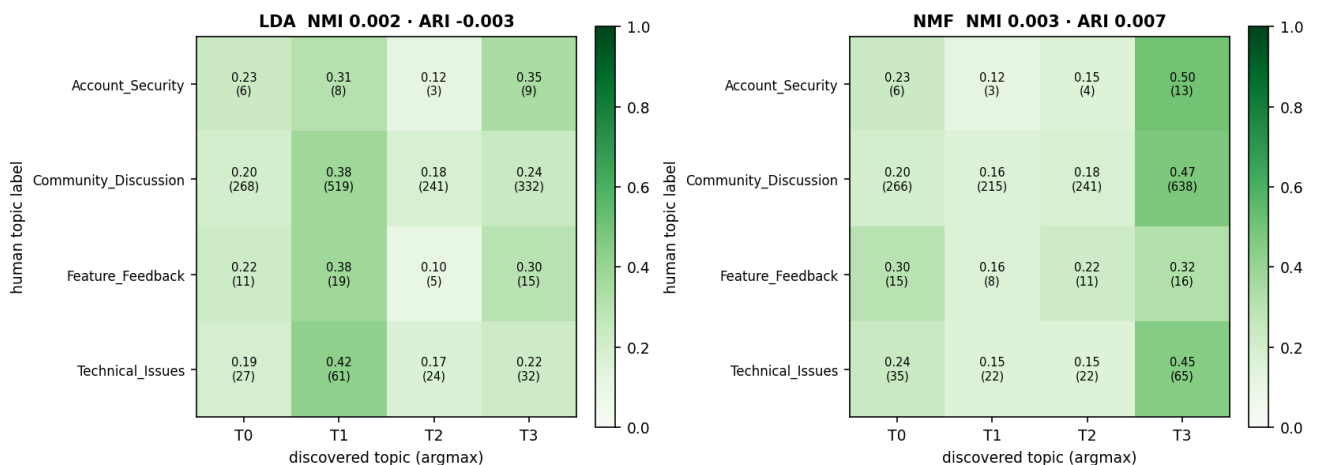


Figure: human topic (rows) vs discovered topic (columns); cell = row share with raw count.

## Discovered topics -- LDA top-10 words (ranked)

### LDA T0

1. <user>
2. <hashtag>
3. time
4. day
5. 3rd
6. 1st
7. sunday
8. mon
9. c
10. thursday

### LDA T2

1. tomorrow
2. game
3. <hashtag>
4. <user>
5. 1st
6. sunday
7. 2
8. 1
9. 2nd
10. day

### LDA T1

1. <user>
2. tomorrow
3. not
4. going
5. friday
6. see
7. go
8. day
9. night
10. make

### LDA T3

1. <hashtag>
2. not
3. <user>
4. night
5. tomorrow
6. day
7. saturday
8. time
9. last
10. thursday

Figure: LDA discovered topics -- top-10 words per topic, ranked by component weight.

## Discovered topics -- NMF top-10 words (ranked)

### NMF T0

1. <user>
2. 1st
3. 2nd
4. u
5. friday
6. rt
7. sunday
8. <smile>
9. no
10. going

### NMF T2

1. not
2. NOT\_wait
3. NOT\_see
4. NOT\_want
5. NOT\_tomorrow
6. NOT\_go
7. NOT\_even
8. NOT\_know
9. NOT\_going
10. 3rd

### NMF T1

1. <hashtag>
2. day
3. 1st
4. game
5. night
6. sunday
7. 1
8. 2
9. new
10. saturday

### NMF T3

1. tomorrow
2. going
3. day
4. see
5. night
6. go
7. game
8. tonight
9. today
10. <smile>

Figure: NMF discovered topics -- top-10 words per topic, ranked by component weight.

## Reproducibility

| component       | version                   |
|-----------------|---------------------------|
| python          | 3.13.14                   |
| sklearn         | 1.6.1                     |
| scipy           | 1.17.1                    |
| pandas          | 2.2.3                     |
| numpy           | 2.3.5                     |
| matplotlib      | 3.10.9                    |
| seed            | 42                        |
| train time (s)  | 273.2                     |
| eval time (s)   | 57.1                      |
| generated (UTC) | 2026-09-18T06:23:43+00:00 |

- Reproduce: `pip install -r requirements-round2.txt && ./run_round2.sh`
- Models: `models/round2/{sentiment_best,topic_best,topic_lda,topic_nmf}.pkl`
- Metrics source: `output/round2/metrics.json` (this PDF renders it)

- Inference: `python src/round2/predict.py --text "..."`